# MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 8

**1.** *Compute the following determinants using* ***row reduction to echelon form****:*

(1) $\begin{vmatrix} 1 & 5 & -4 \\ -1 & -4 & 5 \\ -2 & -8 & 7 \end{vmatrix}$

Recall that using replacement does not change the determinant:

$$\begin{vmatrix} 1 & 5 & -4 \\ -1 & -4 & 5 \\ -2 & -8 & 7 \end{vmatrix} \xrightarrow{R_2 \to R_2 + R_1, R_3 \to R_3 + 2R_1} \begin{vmatrix} 1 & 5 & -4 \\ 0 & 1 & 1 \\ 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{vmatrix} 1 & 5 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{vmatrix} = -3$$

(2) $\begin{vmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{vmatrix}$

Recall that if a matrix has duplicate rows (or columns), then the determinant is zero:

$$\begin{vmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{vmatrix} \xrightarrow{R_2 \to R_2 + 2R_1, R_3 \to R_3 - 3R_1, R_4 \to R_4 - R_1} \begin{vmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & -4 & 2 & -5 \\ 0 & -4 & 2 & -5 \end{vmatrix} = 0$$

**2.** *Find the determinant of $A$ if*

$$A = S \begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix} S^{-1}.$$

$$\begin{aligned} \det(A) &= \det(S \begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix} S^{-1}) = \det(S) \det(\begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix}) \det(S)^{-1} \\ &= \det(\begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix}) \det(S) \det(S)^{-1} = \det(\begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix}) = \lambda_1 \lambda_2. \end{aligned}$$

**3.** *Let $A$ be an $n \times n$-matrix with eigenvalue $\lambda$. Which of the following statements are true? Give reasons why!*

(a) $\lambda^2$ is an eigenvalue of $A^2$.

True. Let $\mathbf{v}$ be an eigenvector of $A$ to the eigenvalue $\lambda$. Then

$$A^2\mathbf{v} = A(A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda A\mathbf{v} = \lambda\lambda\mathbf{v} = \lambda^2\mathbf{v}.$$

So $\mathbf{v}$ is an eigenvector of $A^2$ to eigenvalue $\lambda^2$. So $\lambda^2$ is an eigenvalue of $A^2$.

(b) $\lambda^{-1}$ is an eigenvalue of $A^{-1}$.

True. Let $\mathbf{v}$ be an eigenvector of $A$ to the eigenvalue $\lambda$. Then
$$A^{-1}(\lambda\mathbf{v}) = A^{-1}(A\mathbf{v}) = (A^{-1}A)\mathbf{v} = \mathbf{v} = \lambda^{-1}(\lambda\mathbf{v}).$$
Hence $\lambda\mathbf{v}$ is an eigenvector of $A^{-1}$ to eigenvalue $\lambda^{-1}$. So $\lambda^{-1}$ is an eigenvalue of $A^{-1}$.

(c) $\lambda + 1$ is an eigenvalue of $A + I$.

True. Let $\mathbf{v}$ be an eigenvector of $A$ to the eigenvalue $\lambda$. Then
$$(A + I)(\mathbf{v}) = A\mathbf{v} + \mathbf{v} = \lambda\mathbf{v} + \mathbf{v} = (\lambda + 1)\mathbf{v}.$$
Hence $(\lambda + 1)\mathbf{v}$ is an eigenvector of $A + I$ to eigenvalue $\lambda + 1$. So $\lambda + 1$ is an eigenvalue of $A + I$.

**4.** *True/False. Give reasons why! Below $A$ and $B$ are matrices of an appropriate size.*

(1) Suppose that $A$ and $B$ are both $n \times n$ matrices. Then $\det(A+B) = \det(A) + \det(B)$.

False, Consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = -A$.

(2) Suppose that $A$ is a $3 \times 3$ matrix with $\det(A) = 9$. Then $\det(2A) = 18$.

False, $\det(2A) = 2^3 \det(A) = 72$.

(3) Suppose that $A$ is a $2 \times 3$ matrix. Then $\det(A^TA) = 0$.

True. Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$. Write $\boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Then
$$A^TA = \begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = \begin{bmatrix} (a_1\boldsymbol{a} + b_1\boldsymbol{b}) & (a_2\boldsymbol{a} + b_2\boldsymbol{b}) & (a_3\boldsymbol{a} + b_3\boldsymbol{b}) \end{bmatrix}.$$
Therefore the columns of $A^TA$ are in $\operatorname{span}(\boldsymbol{a}, \boldsymbol{b})$. Since $A^TA$ has three columns, they have to be linearly dependent. Hence $A^TA$ is not invertible and $\det(A^TA) = 0$.

(4) Suppose that $A$ is a $2 \times 3$ matrix. Then $\det(AA^T) = 0$.

False. Consider
$$A = \begin{pmatrix} 1 & 2 & -5 \\ 3 & -1 & 4 \end{pmatrix}.$$
Then
$$AA^T = \begin{pmatrix} 30 & -19 \\ -19 & 26 \end{pmatrix} = B$$
so that $\det B = 419 \neq 0$.

(5) The determinant of $A$ is the product of the pivots in any echelon form $U$ of $A$, multiplied by $(-1)^r$, where $r$ is the number of row interchanges made during row reduction from $A$ to $U$.

Consider $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the RREF of $A$. $A$ can be brought to $U$ using no row exchanges. However $\det(A) = 4$ whereas $\det(U) = 1$.

(6) A square matrix $A$ is not invertible iff 0 is an eigenvalue of $A$.

True. It follows from the fact that the determinant is the product of eigenvalues.

(7) The eigenvalues of a square matrix are the numbers on its main diagonal.

False, see for example 3(1) and 3(4) above.

(8) If $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent eigenvectors of a square matrix $A$, then they correspond to different eigenvalues.

False, consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. $A$ has two linearly independent eigenvectors (e.g., any basis of $\mathbb{R}^2$). However, all eigenvectors correspond to the same eigenvalue, $\lambda = 1$.

(9) The sum of two eigenvectors of a matrix $A$ is also an eigenvector of $A$.

False. Consider $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The sum of these two eigenvectors is not an eigenvector.

**5.** *Let $A$ be a $2 \times 2$-matrix , and suppose that $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of $A$ with eigenvalue* 0 *and $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ is an eigenvector of $A$ with eigenvalue 1. Determine $A$.*

Suppose that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. By the definition of eigenvectors, we have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Solving these equations, we obtain that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix}.$$

**6.** *Let $A$ be an $n \times n$-matrix and let $\lambda_1, \lambda_2$ be real numbers such that $\lambda_1 \neq \lambda_2$. Let $\mathbf{x}, \mathbf{y}$ be such that $A\mathbf{x} = \lambda_1 \mathbf{x}$ and $A^T\mathbf{y} = \lambda_2 \mathbf{y}$. Check that $\mathbf{x}$ and $\mathbf{y}$ are orthogonal to each other.*

Note that

$$\lambda_1 \mathbf{x}^T \mathbf{y} = (A\mathbf{x})^T \mathbf{y} = \mathbf{x}^T (A^T \mathbf{y}) = \mathbf{x}^T (\lambda_2 \mathbf{y}) = \lambda_2 \mathbf{x}^T \mathbf{y}.$$

Since $\lambda_1 \neq \lambda_2$, we must have $\mathbf{x}^T \mathbf{y} = 0$. i.e. $\mathbf{x}$ and $\mathbf{y}$ are orthogonal to each other.

**7.** *This exercise explores a particular type of matrices, called* ***nilpotent matrices****, which have interesting properties and they are an useful source of counterexamples in mathematics! A square matrix $A$ is nilpotent if there exists a positive integer $k$ such that $A^k = 0$ (i.e. the $k$-th power of the matrix $A$ is the zero matrix).*

(a) Give two examples of non zero nilpotent $3 \times 3$ matrices.

Of course the $3 \times 3$ matrix made of zeros is nilpotent. A simple $2 \times 2$ example would be

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

For $3 \times 3$, take for instance

$$B = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

but less trivial ones are given by

$$C = \begin{pmatrix} 5 & -3 & 2 \\ 15 & -9 & 6 \\ 10 & -6 & 4 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \\ -1 & -2 & -5 \end{pmatrix}.$$

(b) Explain why a nilpotent matrix is non invertible. (Hint: Use determinants).

Let $A$ be a nilpotent matrix, i.e. $A^n = 0$. It follows from the fact that $\det(A^k) = (\det(A))^k = 0$, then $\det A = 0$ therefore $A$ is non invertible.

(c) Explain why all the eigenvalues of a nilpotent matrix $A$ are 0.

Let $A$ be a nilpotent matrix, i.e. $A^n = 0$. Let $\lambda$ be an eigenvalue of $A$ so that $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero vector $\mathbf{x}$. We then have that $\lambda^n\mathbf{x} = A^n\mathbf{x} = 0$ so that $\lambda = 0$.

**8.** *Let $A, B$ be two $n \times n$-matrices such that $AB = BA$.*

(a) Suppose $\boldsymbol{v}$ is an eigenvector of $A$ with eigenvalue $\lambda$. Is $B\boldsymbol{v}$ an eigenvector of $A$? If so, what is the eigenvalue of that eigenvector?

We first must consider the case that $B\boldsymbol{v} = 0$, in which case $B\boldsymbol{v}$ cannot be an eigenvector. In the other case, consider $A(B\boldsymbol{v}) = (AB)\boldsymbol{v}$, and since $AB = BA$, this is the same as $(BA)\boldsymbol{v} = B(A\boldsymbol{v}) = B(\lambda\boldsymbol{v}) = \lambda(B\boldsymbol{v})$ (as $\boldsymbol{v}$ is an eigenvector of $A$ with eigenvalue $\lambda$), so since $A(B\boldsymbol{v}) = \lambda B\boldsymbol{v}$, $B\boldsymbol{v}$ is an eigenvector of $A$ with eigenvalue $\lambda$. So overall either $B\boldsymbol{v} = 0$ or $B\boldsymbol{v}$ is an eigenvector of $A$ with eigenvalue $\lambda_i$.

(b) Suppose $A$ has eigenvectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ with distinct eigenvalues $\lambda_1 \neq ... \neq \lambda_n$. Is each $\boldsymbol{v_i}$ also an eigenvector of $B$? (This question is a bit tricker. Hint: Note that each of the eigenspaces of $A$ has dimension 1 and then use your answer to (a).).

Take the $i$-th eigenvalue $\boldsymbol{v_i}$ of $A$. Since there are $n$ distinct eigenvalue, $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ form a basis of $\mathbb{R}^n$. Thus the eigenspace of $A$ for eigenvalue $\lambda_i$ is span($\boldsymbol{v_i}$). Since $\boldsymbol{v_i}$ is an eigenvector of $A$ with eigenvalue $\lambda_i$, we get $A\boldsymbol{v_i} = \lambda_i \boldsymbol{v_i}$. Then by **a.** either $B\boldsymbol{v_i} = 0$ or $B\boldsymbol{v_i}$ is an eigenvector of $A$ to the eigenvalue $\lambda_i$. If $B\boldsymbol{v_i} = 0$, then $\boldsymbol{v_i}$ is eigenvector of $B$ with eigenvalue 0. So it is left to consider the case that $B\boldsymbol{v_i}$ is an eigenvector of $A$ to the eigenvalue $\lambda_i$. Then $B\boldsymbol{v_i}$ is in the eigenspace of $A$ of the eigenvalue $\lambda_i$. Then $B\boldsymbol{v_i}$ is in the span of $\boldsymbol{v_i}$. Therefore $B\boldsymbol{v_i}$ is a multiple of $\boldsymbol{v}_i$. Hence $\boldsymbol{v_i}$ is an eigenvector of $B$.